Predicting Osteoporosis (Bone Fracture) Risk in Women

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Data Introduction & Objective Summary

Data Description: This dataset contains clinical, demographic, and treatment information including age, BMI, prior and family fracture history, fracture-risk scores, rater and site identifiers, and bone-medication exposure which is used to model incident fractures and evaluate effects of bone-active therapies.

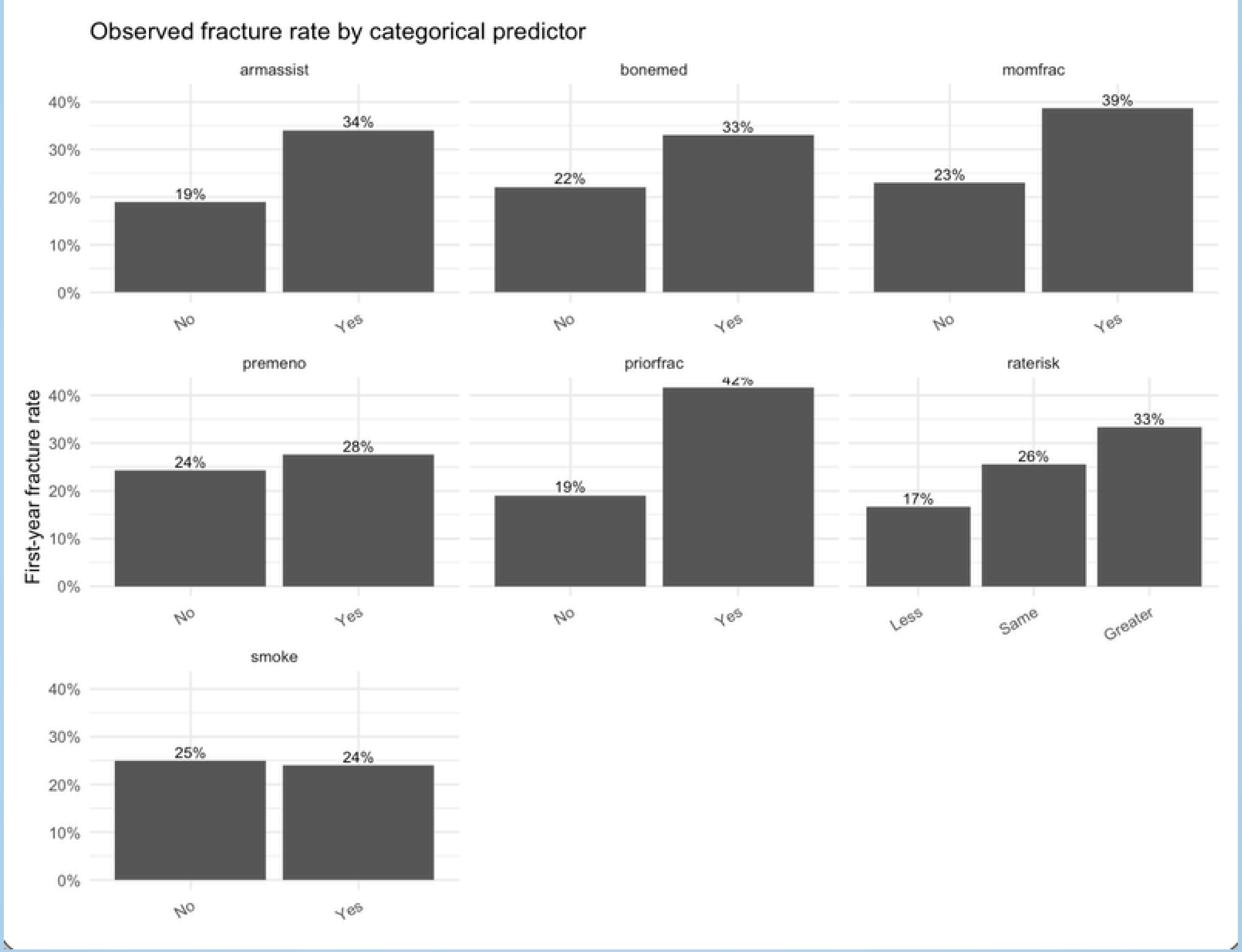
Objective 1: Perform EDA on training set using a 75/25 split and use intuition and analysis to build out a simple MLR model.

Objective 2: Building and comparing multiple predictive classification models, using your simple logistic regression from Objective 1 as the baseline.

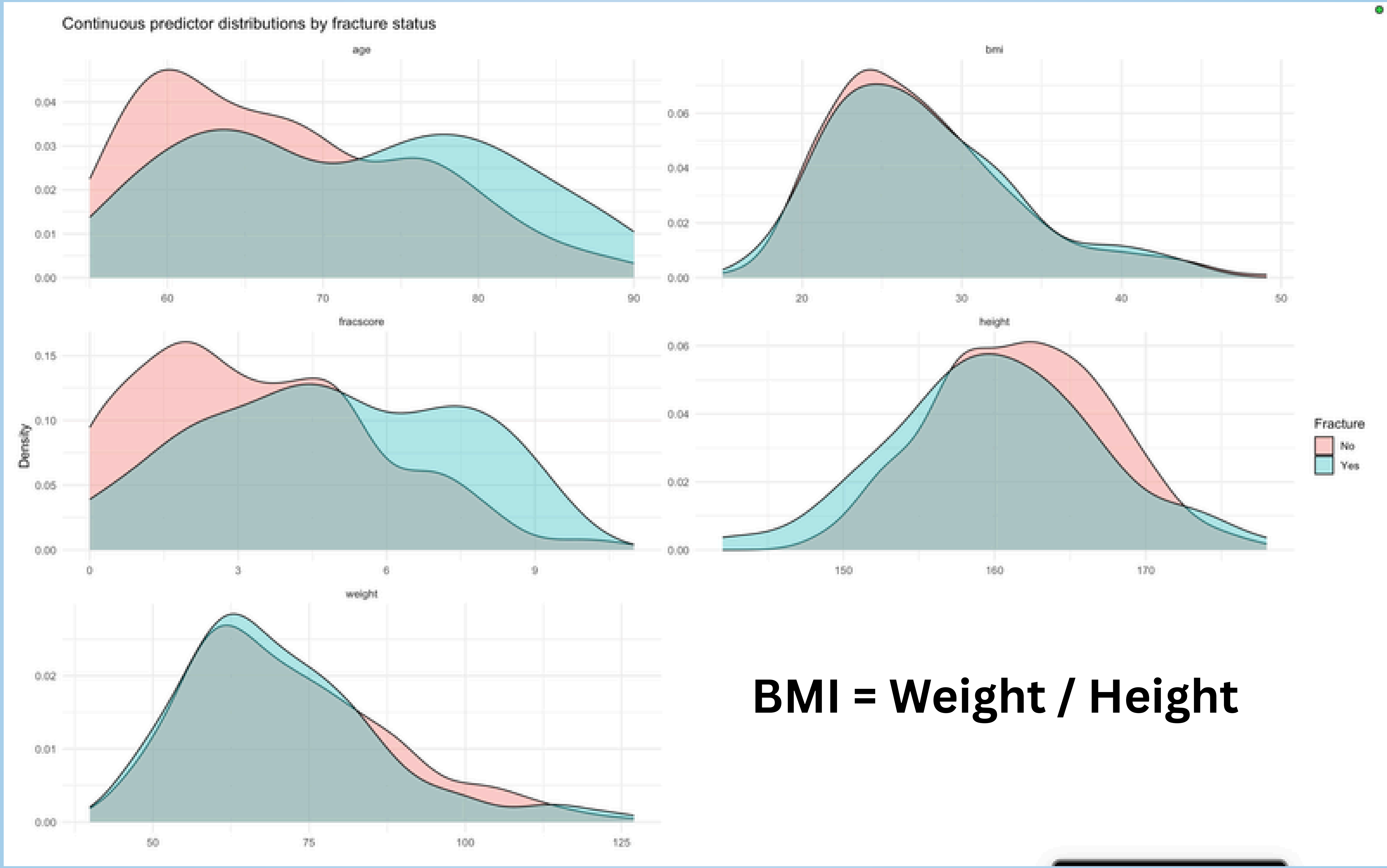
Summary of Variables

Variable	Type	Description
sub_id	integer	Participant identifier
site_id	integer	Study site identifier
phy_id	integer	Physician or rater identifier
priorfrac	factor	Prior fracture history
age	integer	Age at baseline
weight	numeric	Body weight
height	integer	Height
bmi	numeric	Body mass index
premen o *(premen o)*	factor	Menopausal status
momfrac	factor	Mom had a fracture
armassist	factor	Arm assistance or functional status
smoke	factor	Smoking status
raterisk	factor	Risk category assigned by rater
fracscore	integer	Fracture risk score
fracture	factor	Fracture outcome at follow-up
bonemed	factor	Bone medications at enrollment
bonemed_fu	factor	Bone medication use during follow-up
bonetreat	factor	Bone medications both at enrollment and follow-up

Exploratory Data Analysis

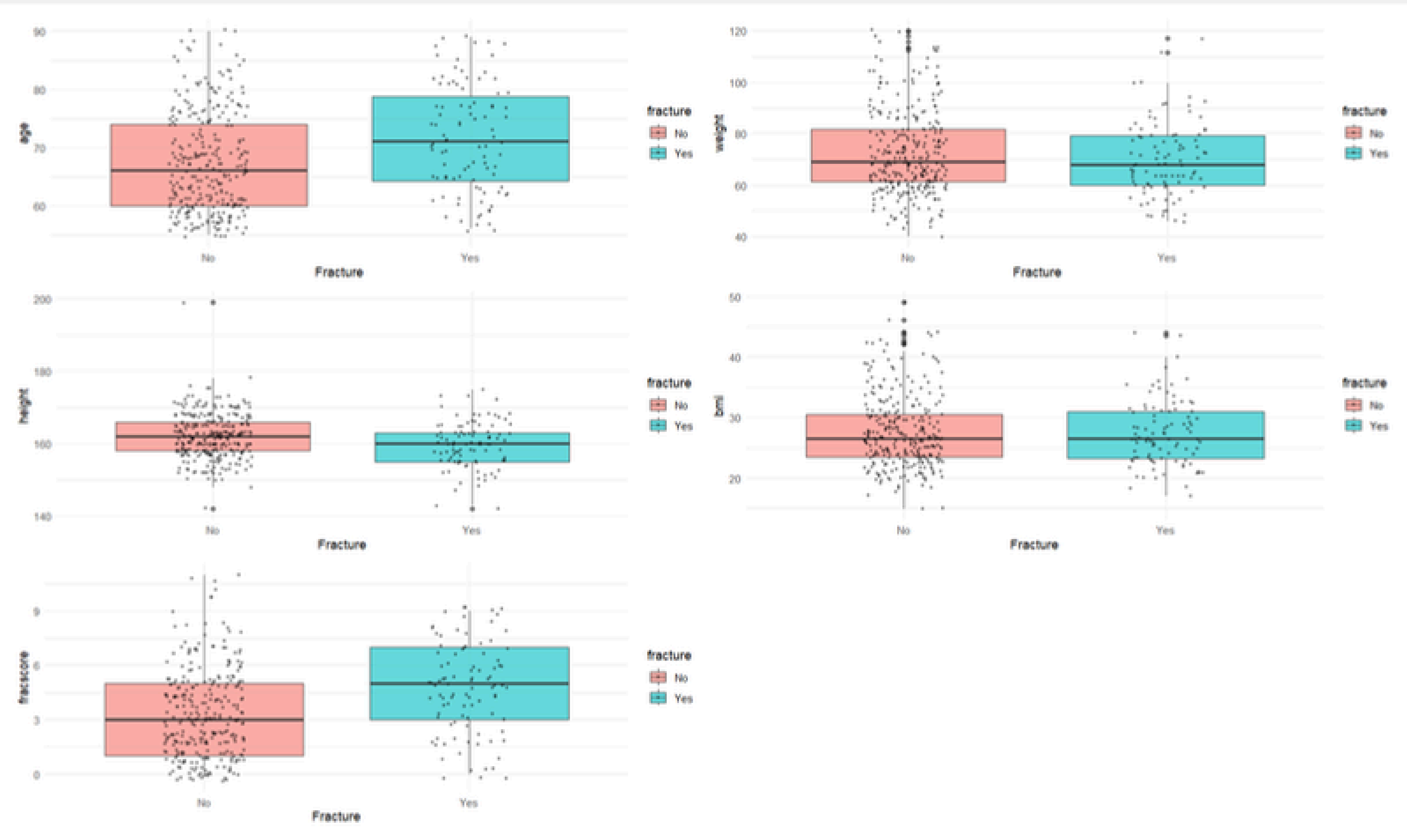


Exploratory Data Analysis

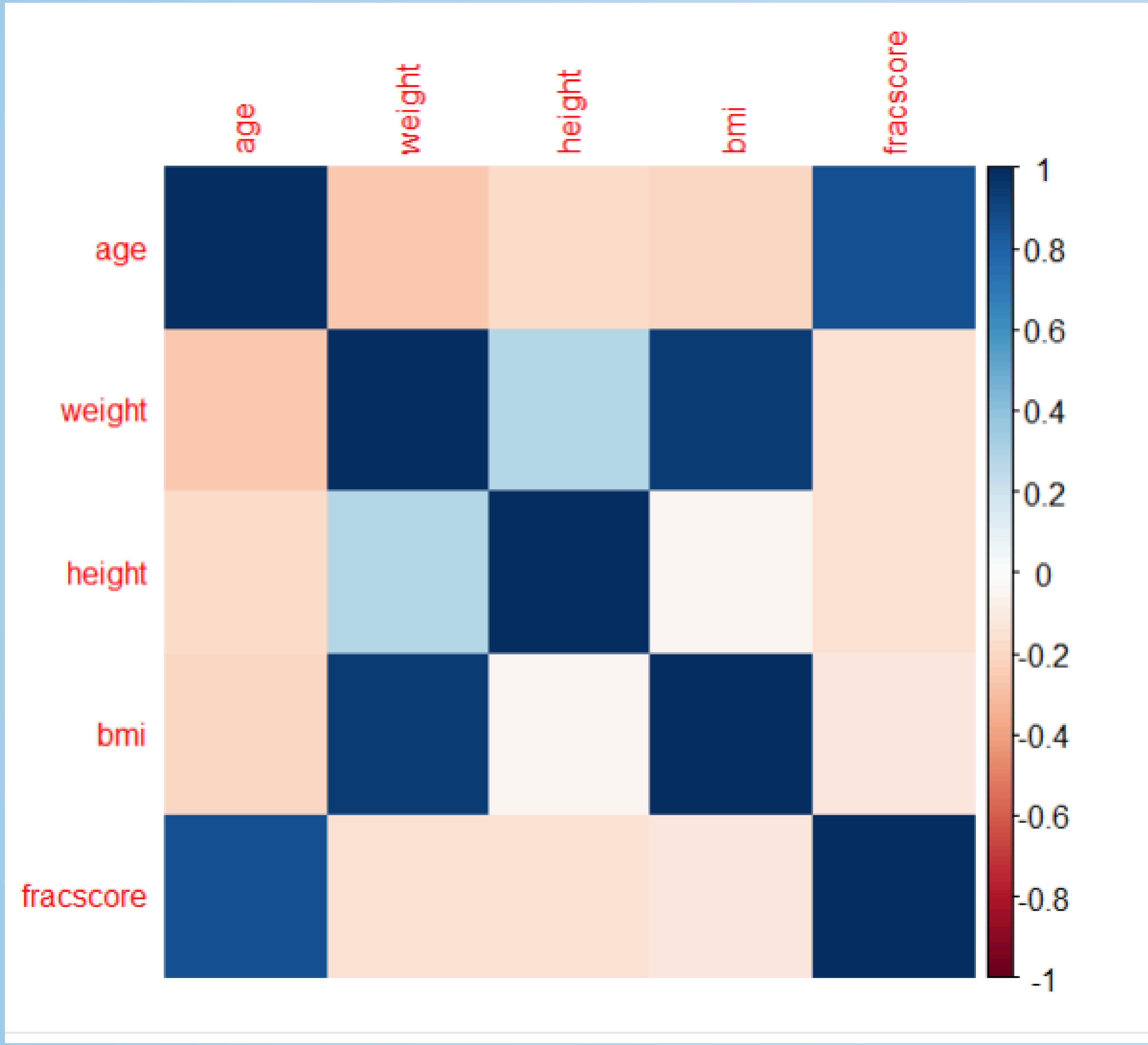


BMI = Weight / Height

Exploratory Data Analysis



Checking Correlation for Evidence of Multicollinarity



Age and Fracscore Highly Correlated

Because Fracscore is questionable to how that variable came about we would rather choose to keep age because of its practicality.

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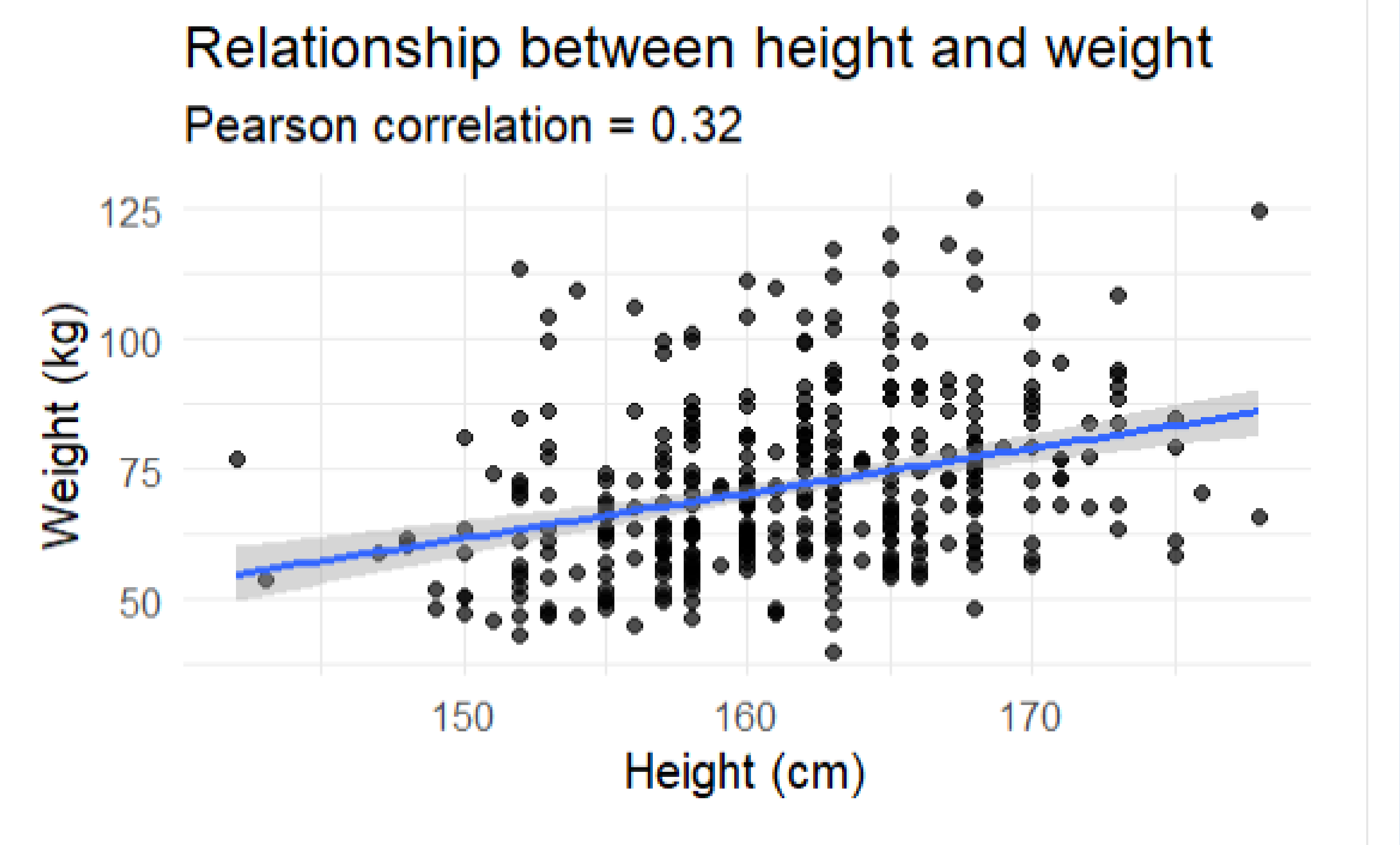
Height and Weight Slightly Correlated

BMI and Weight Highly Correlated

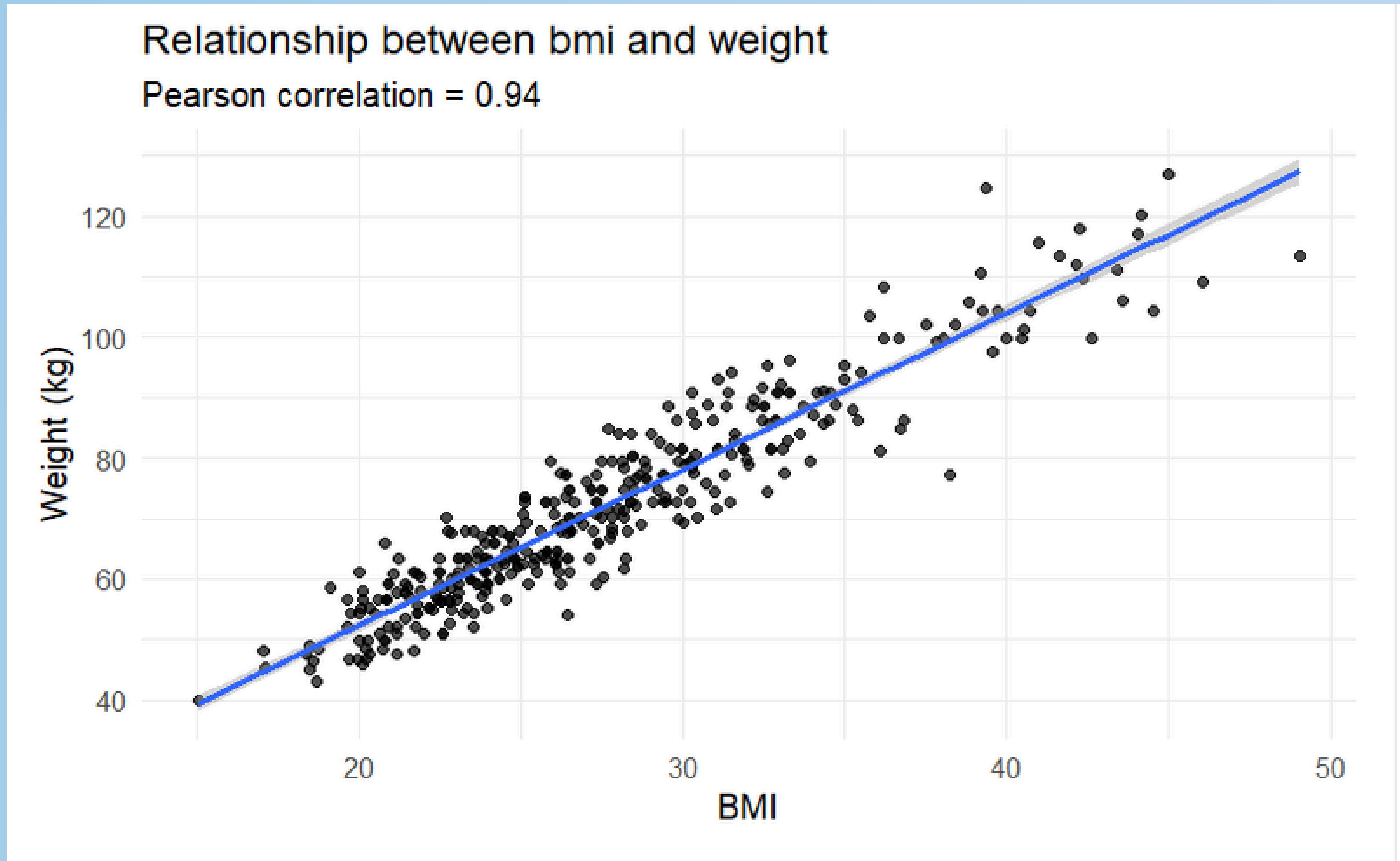
Checking Multicolinarity between Height and Weight

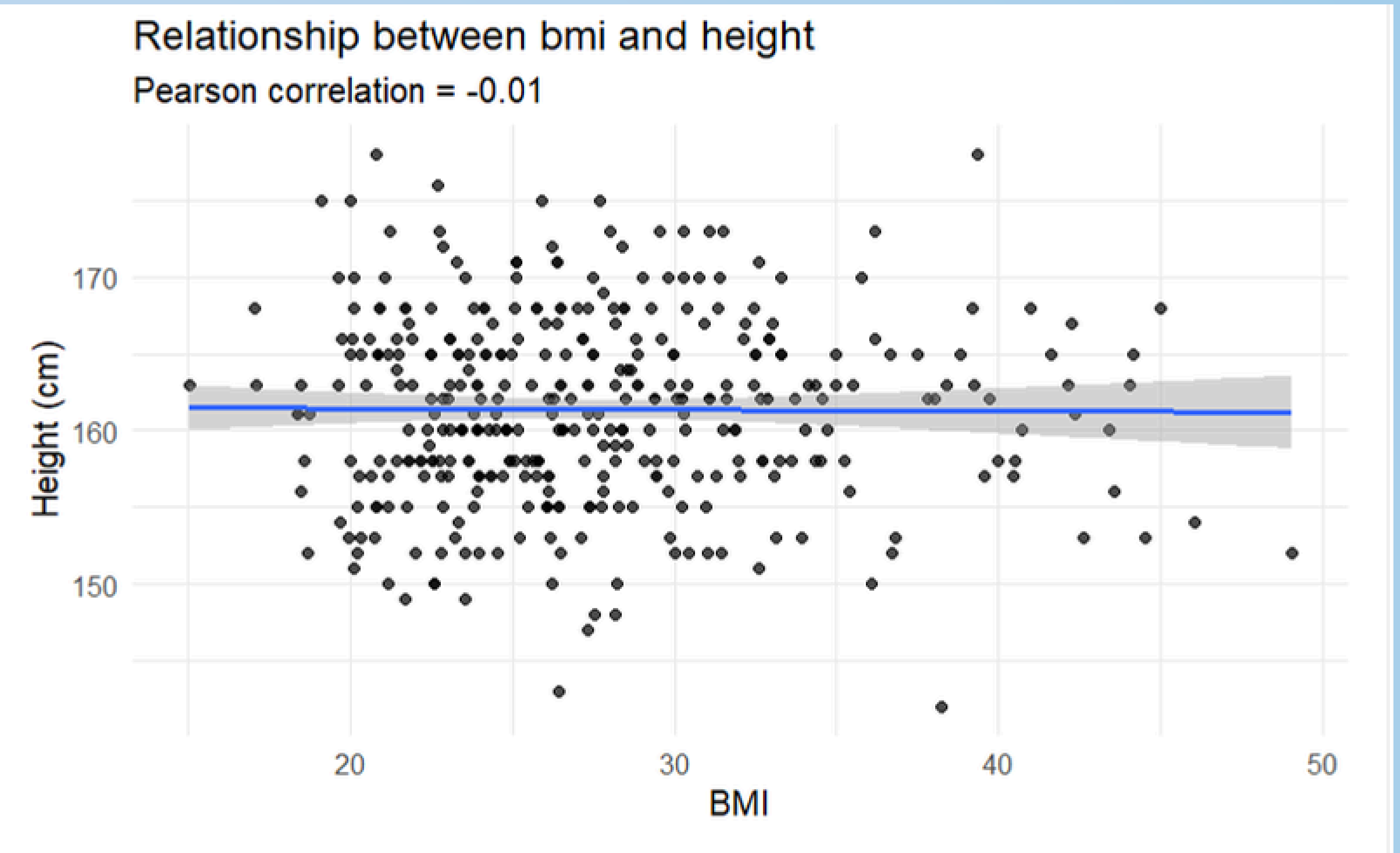


	GVIF	Df	GVIF^(1/(2*Df))
age	1.480790	1	1.216877
height	1.202903	1	1.096769
weight	1.641076	1	1.281045
priorfrac	1.162599	1	1.078239
momfrac	1.093152	1	1.045539
armassist	1.383808	1	1.176354
smoke	1.076319	1	1.037458
premeno	1.123716	1	1.060055
raterisk	1.260331	2	1.059549
bonemed	1.252324	1	1.119073

Height and weight had a modest Pearson correlation of 0.32. Adjusted generalized variance inflation factors were close to 1 for all predictors, indicating no problematic multicollinearity between the two variables we suspected. Therefore, height and weight were retained for together for further investigation.

Checking Multicolinarity between Height and Weight + BMI





	GVIF	Df	GVIF^(1/(2*Df))
age	1.485741	1	1.218910
bmi	173.381323	1	13.167434
height	21.681315	1	4.656320
weight	191.256048	1	13.829535
priorfrac	1.171380	1	1.082303
momfrac	1.096505	1	1.047141
armassist	1.380470	1	1.174934
smoke	1.083186	1	1.040762
premeno	1.142395	1	1.068829
raterisk	1.261662	2	1.059829
bonemed	1.247027	1	1.116704

In logistic regression, GVIF-adjusted values above 2–5 typically indicate meaningful multicollinearity. Values above 10 are considered severe. Here, both BMI and weight exceed 13, and height exceeds 4.5, clearly signaling problematic collinearity with having both weight or height and continuous bmi in the model.

Exploring Solution to Keep BMI in another Form

Here we are looking to explore whether we should either exclude BMI in the model or see if there is another form we could use to keep the predictor as it is an important measure in clinical diagnosis.

By categorizing BMI removes the linearity assumption and collinearity issues with weight and height while aligning with clinically meaningful thresholds (underweight, normal, overweight, obesity). These categories are widely used in fracture-risk guidelines and improve interpretability of odds ratios.

Model where BMI is continous

$$\text{fracture} \sim \beta_0 + \beta_1(\text{age}) + \beta_2(\text{bmi}) + \beta_3(\text{priorfrac}) + \beta_4(\text{momfrac}) + \beta_5(\text{armassist}) + \beta_6(\text{smoke}) + \beta_8(\text{premeno}) + \beta_8(\text{raterisk}) + \beta_9(\text{bonemed})$$

VS

Model where BMI is categorcial

$$\text{fracture} \sim \beta_0 + \beta_1(\text{age}) + \beta_2(\text{bmi\_categorical}) + \beta_3(\text{priorfrac}) + \beta_4(\text{momfrac}) + \beta_5(\text{armassist}) + \beta_6(\text{smoke}) + \beta_8(\text{premeno}) + \beta_8(\text{raterisk}) + \beta_9(\text{bonemed})$$

	df	AIC
	<dbl>	<dbl>
model_cont_bmi	11	401.1629
model_cat_bmi	13	403.7469

Because Objective 1 emphasizes interpretability rather than predictive optimization, the categorical BMI model is preferable even if its AIC is marginally higher.

	df	AIC
	<dbl>	<dbl>
model_cat_bmi	13	403.7469
model_cat_bmi_height_weight	15	403.1191
model_cat_bmi_weight	14	402.5967
model_cat_bmi_height	14	401.7003

Because BMI is now categorical, height and weight provide additional nuance without being linearly redundant. Among the candidate models, the combination of categorical BMI and height produced the lowest AIC, making it the most interpretable and best-performing additive model.

	df	AIC
	<dbl>	<dbl>
model_cat_bmi_height	14	401.7003
model_height_weight	12	399.1812
model_height_only	11	397.6989
model_weight_only	11	402.1198

The model using categorical BMI plus height provides the best balance of clinical interpretability and performance, so we retain the categorical BMI predictor.

EDA Strongest Candidate Model for Interpretability

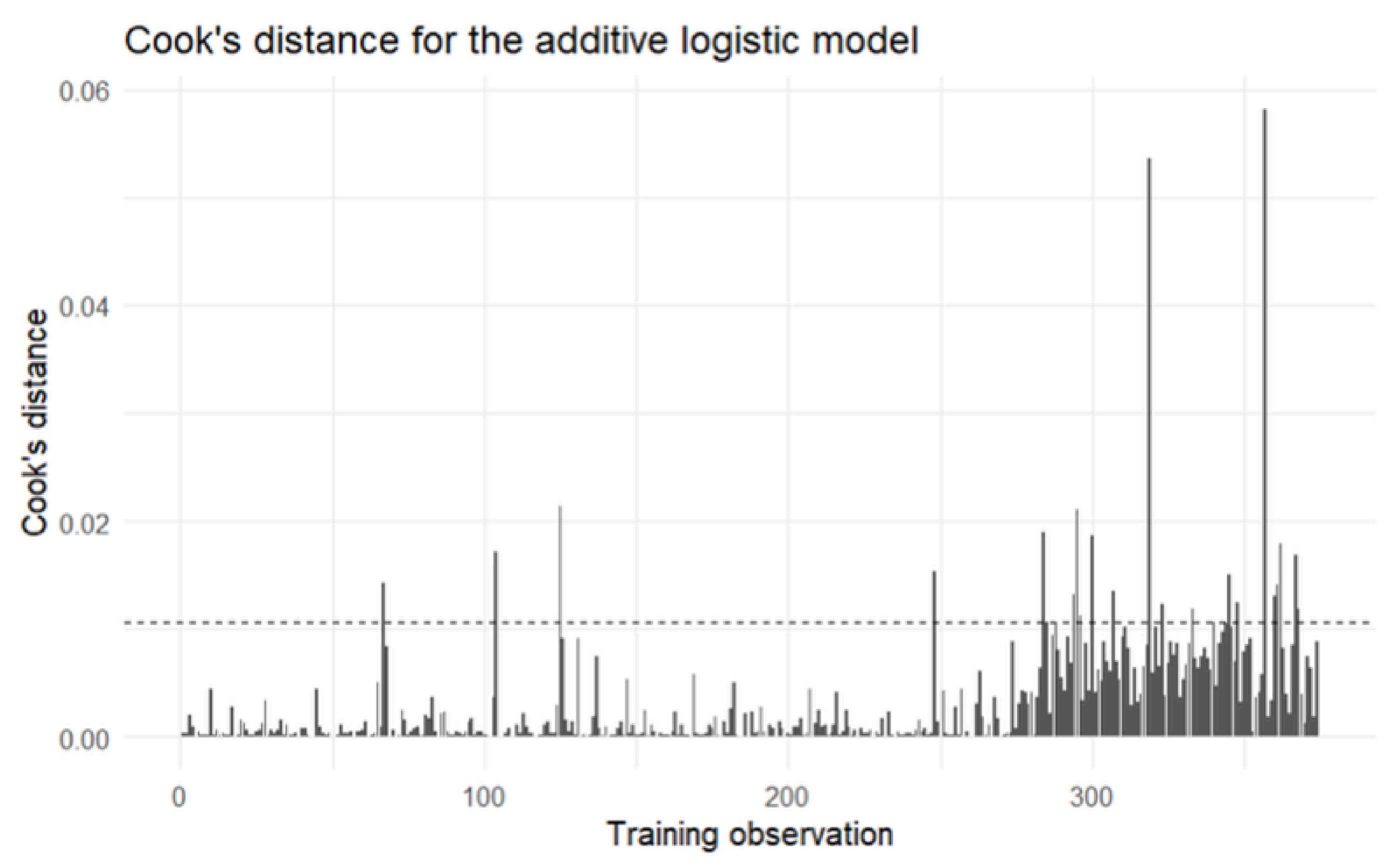
Categorical BMI with Height

fracture ~ β_0 + β_1(age) + β_2(bmi_categorical) + β_3(height) + β_4(priorfrac) + β_5(momfrac) + β_6(armassist) + β_7(smoke) + β_8(premeno) + β_9(raterisk) + β_{10}(bonemed)

Additive Model with Height for Objective 2

fracture ~ β0 + β1(age) + β2(height) + β3(priorfrac) + β4 (momfrac) + β5(armassist) + β6(smoke) + β7(premeno) + β8(raterisk) + β9(bonemed)

Residual Diagnostics



Cook's d: Several observations were flagged as potentially influential using the 4/n screening rule, but none had an exceptionally large Cook's distance. These observations were retained and examined through sensitivity analysis.

observation <int>	fracture <fctr>	.fitted <dbl>	.std.resid <dbl>
292	Yes	-2.502286	2.283002
374	Yes	-2.485155	2.275703
344	Yes	-2.402925	2.244853
333	Yes	-2.291425	2.202000
327	Yes	-2.280782	2.191669
366	Yes	-2.099351	2.118885
318	Yes	-2.092238	2.115856
323	Yes	-2.045249	2.102804
294	Yes	-1.890446	2.040852
357	Yes	-1.595872	2.010435

Standardized Residuals: Nine observations had standardized residuals greater than 2 and all nine were fractured cases. Because .fitted values are reported on the log-odds scale, these large residuals are expected for observed events with low predicted log-odds.

Interpretation of Objective 1 Model

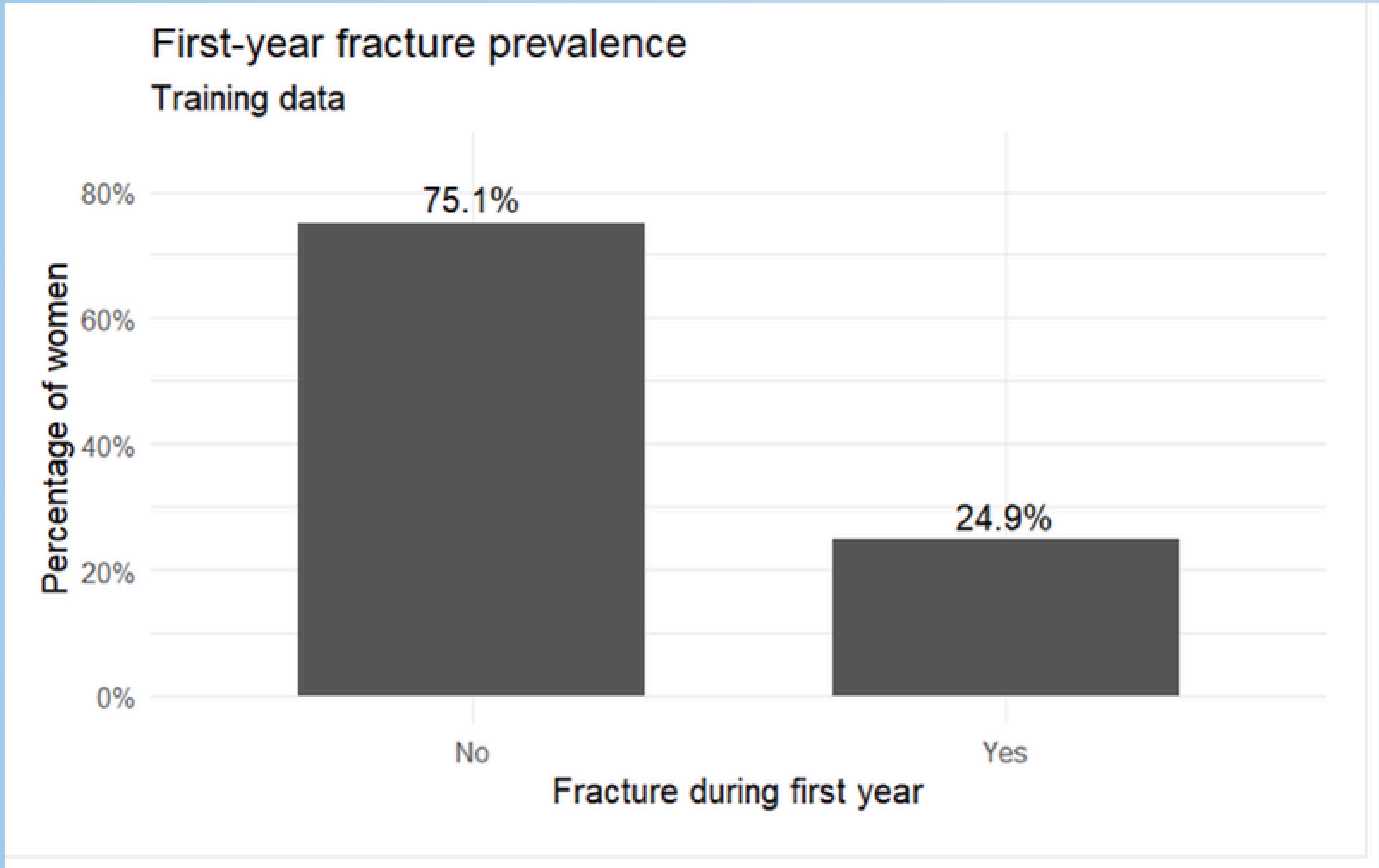
```
Call:
glm(formula = fracture ~ age + height + bmi_categorical + priorfrac +
    momfrac + armassist + smoke + raterisk + bonemed, family = binomial(link =
"logit"),
    data = bone_train)

Coefficients:
                            Estimate Std. Error z value Pr(>|z|)
(Intercept)                  2.50181    3.85164   0.650  0.51599
age                          0.03603    0.01638   2.199  0.02785 *
height                      -0.04459    0.02180  -2.046  0.04080 *
bmi_categoricalUnderweight -0.66009    0.91109  -0.725  0.46875
bmi_categoricalOverweight    0.10097    0.32180   0.314  0.75371
bmi_categoricalObesity       0.40486    0.36276   1.116  0.26439
priorfracYes                 0.77257    0.29143   2.651  0.00803 **
momfracYes                   0.83933    0.37901   2.215  0.02679 *
armassistYes                 0.43638    0.29099   1.500  0.13372
smokeYes                    -0.06788    0.52304  -0.130  0.89675
rateriskSame                 0.34298    0.33510   1.024  0.30606
rateriskGreater              0.64755    0.35932   1.802  0.07152 .
bonemedYes                   0.18294    0.31107   0.588  0.55646
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Older age, shorter height, prior fracture, and maternal fracture history were linked to higher first-year fracture risk. A 10-year increase in age raised the odds by about 43%, while a 10-cm increase in height lowered the odds by about 36–40%. Prior fracture and maternal fracture history were the strongest predictors, each more than doubling the odds of fracture

term <chr>	odds_ratio <dbl>	CI_low <dbl>	CI_high <dbl>	p_value <dbl>
(Intercept)	12.205	0.006	24225.108	0.516
age	1.037	1.004	1.071	0.028
height	0.956	0.916	0.998	0.041
bmi_categoricalUnderweight	0.517	0.067	2.816	0.469
bmi_categoricalOverweight	1.106	0.587	2.081	0.754
bmi_categoricalObesity	1.499	0.736	3.064	0.264
priorfracYes	2.165	1.219	3.833	0.008
momfracYes	2.315	1.089	4.850	0.027
armassistYes	1.547	0.873	2.739	0.134
smokeYes	0.934	0.311	2.487	0.897

BMI category was not statistically significant after accounting for stronger predictors such as age, height, and fracture history. However, BMI categories are still clinically useful because they are familiar, easy to interpret, and help communicate body-size-related risk.

Objective 2: Improving Predictive Performance Metrics



One-quarter of the women experience a fracture during the 1st year. Because fractures are less common than non-fractures, sensitivity may eventually be more informative than overall accuracy when evaluating prediction models.

Because fractures are the adverse event you care about preventing, missing a fracture (a false negative) is more costly than incorrectly flagging someone who won't fracture (a false positive). Sensitivity directly measures how well the model avoids those dangerous false negatives.

A model can achieve high overall accuracy simply by predicting "No fracture" for everyone, while failing to identify the women at actual risk. For this reason, sensitivity, the ability to correctly detect fractures, is a more meaningful performance metric than accuracy in this clinical context.

Complex Logisitc Regression Model Performance

Initial validation comparison at threshold = 0.50

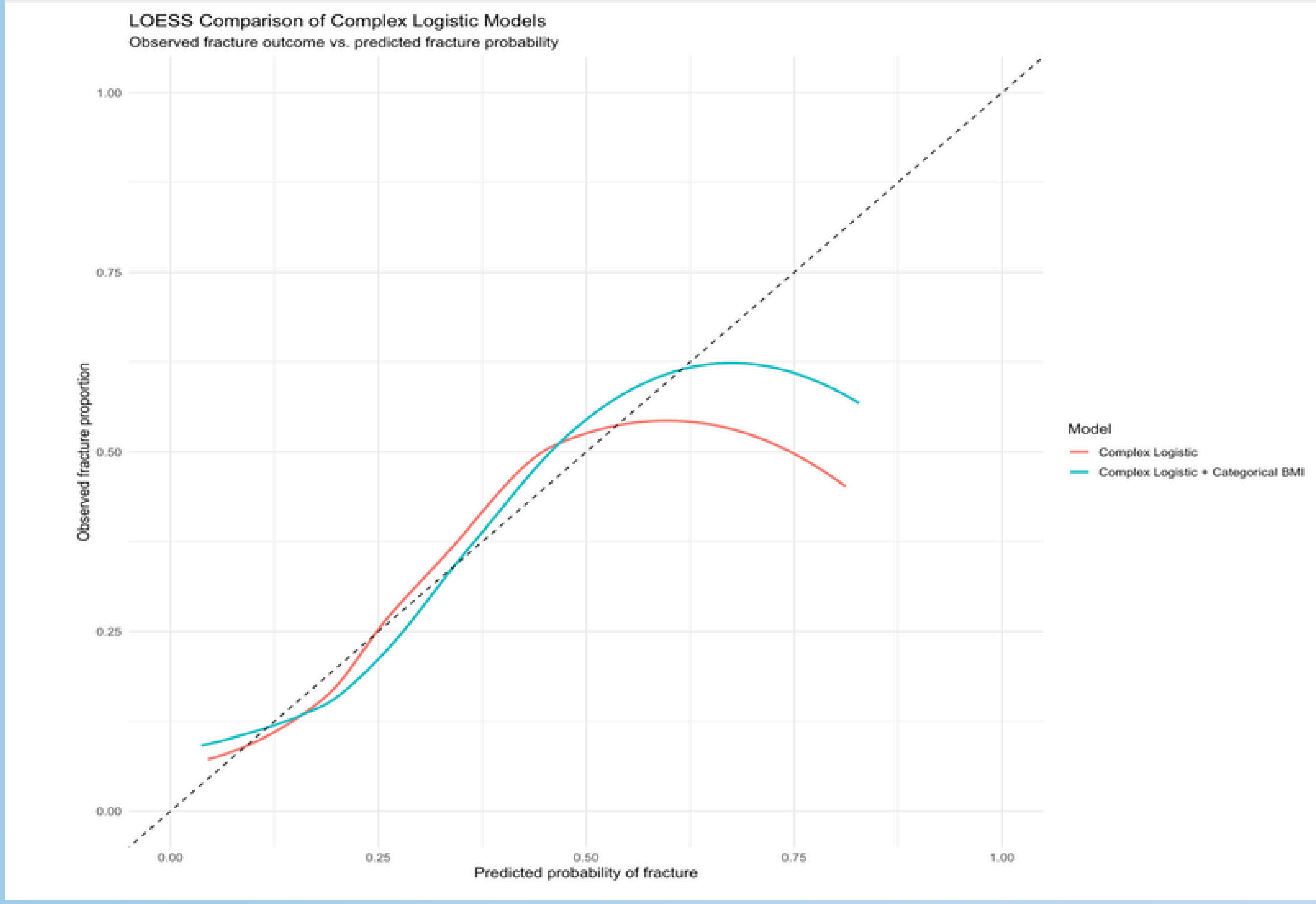
Metric	Additive Model	Complex Logistic Model
Sensitivity	0.219	0.188
Specificity	0.919	0.955
PPV	0.514	0.6
NPV	0.776	0.776
AUROC	0.733	0.718

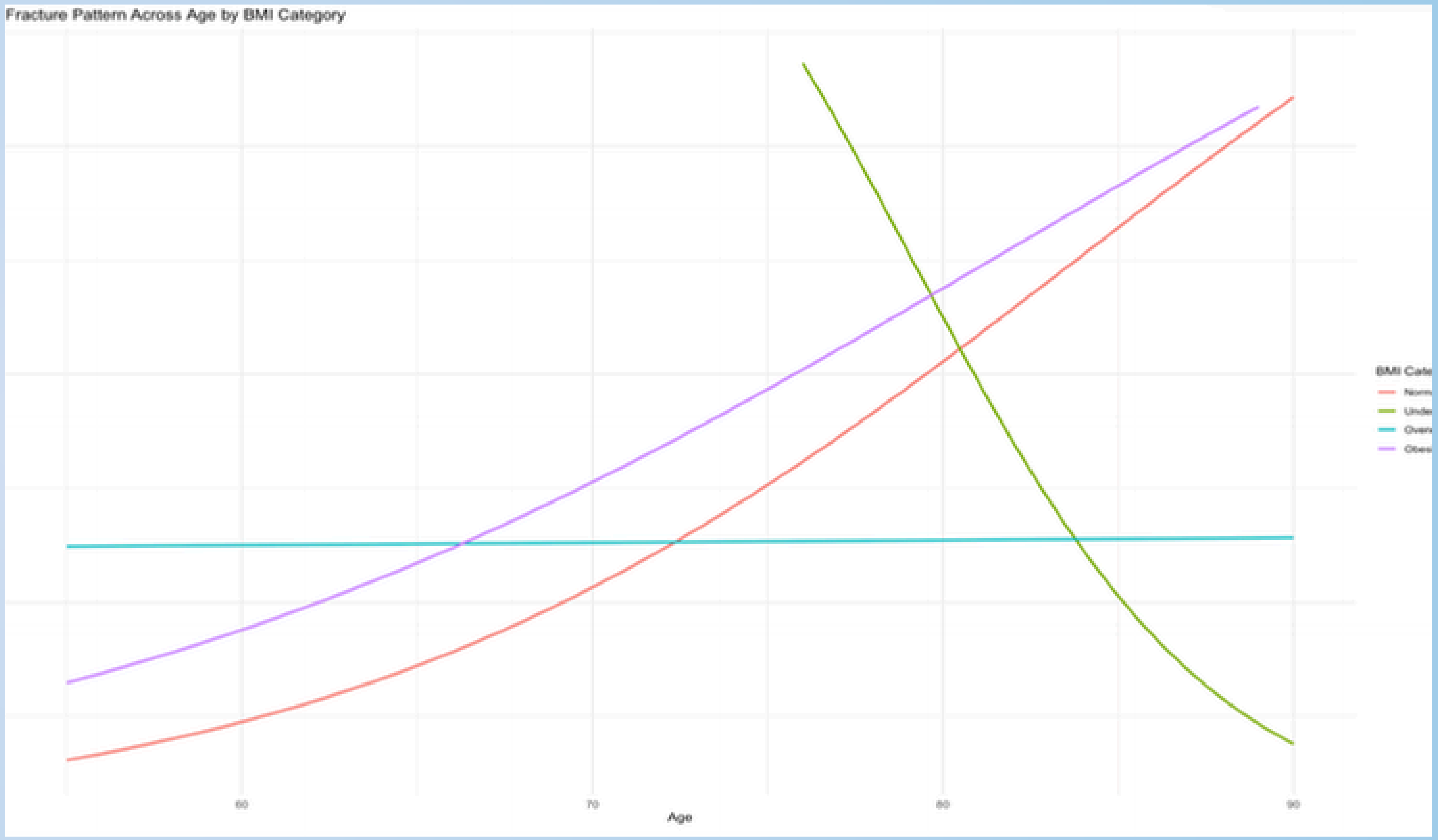
At the default 0.50 threshold, the complex logistic model had lower sensitivity at 18.8%, slightly higher specificity at 95.7%, and a slightly lower AUROC of 0.718.

Overall, the additional complexity did not improve predictive discrimination. Although specificity increased slightly, the complex model identified fewer true fracture cases and had a lower AUROC than the additive model.

This was consistent with the training results, where the complex model also had a higher AIC, indicating that the added complexity was not strongly supported.

Exploring Age x BMI Category Interactions in the Complex Logisitc Model





LDA Model Performance

Initial validation performance at threshold = 0.50

Metric	Additive Model	Complex Logistic Model	LDA Model
Sensitivity	0.219	0.188	0.219
Specificity	0.936	0.952	0.915
PPV	0.538	0.6	0.467
NPV	0.779	0.776	0.775
AUROC	0.723	0.718	0.721

- **LDA was used as another classification method to separate women who fractured from those who did not. At the 0.50 threshold, it had low sensitivity (21.9%) but high specificity (91.5%), with an AUROC of 0.721.**

- **Its AUROC was almost identical to the additive logistic model (0.723), so overall discrimination was very similar. However, the low sensitivity meant many true fracture cases were missed, so a lower threshold was considered later.**

- **LDA also assumes approximate multivariate normality and similar covariance structures between groups. These assumptions were only partly met because the dataset included categorical predictors and non-normal continuous variables.**

QDA Model Performance

Initial validation performance at threshold = 0.50

Model	Threshold	Sensitivity	Specificity	PPV	NPV	Prevalence	AUROC
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Additive Logistic	0.5	0.21875	0.9361702	0.5384615	0.7787611	0.2539683	0.7232380
Complex Logistic	0.5	0.18750	0.9574468	0.6000000	0.7758621	0.2539683	0.7175864
LDA	0.5	0.21875	0.9148936	0.4666667	0.7747748	0.2539683	0.7205785
QDA	0.5	0.34375	0.8510638	0.4400000	0.7920792	0.2539683	0.7145944

- QDA was evaluated as a more flexible alternative to LDA because it allows the covariance structure to differ between fracture groups. At the 0.50 threshold, QDA had higher sensitivity at 34.4%, but lower specificity at 85.1% and PPV at 44%.
- Its AUROC was 0.715, slightly below both additive logistic regression (0.723) and LDA (0.721). So QDA detected more fracture cases, but overall discrimination was not better.
- QDA's assumptions were only partly met because the data included both categorical and non-normal continuous predictors. Overall, QDA did not outperform the simpler additive logistic model.

Random Forest Model Performance

Initial validation performance at threshold = 0.50

Model <chr>	Threshold <dbl>	Sensitivity <dbl>	Specificity <dbl>	PPV <dbl>	NPV <dbl>	Prevalence <dbl>	AUROC <dbl>
Additive Logistic	0.5	0.21875	0.9361702	0.5384615	0.7787611	0.2539683	0.7232380
Complex Logistic	0.5	0.18750	0.9574468	0.6000000	0.7758621	0.2539683	0.7175864
LDA	0.5	0.21875	0.9148936	0.4666667	0.7747748	0.2539683	0.7205785
QDA	0.5	0.34375	0.8510638	0.4400000	0.7920792	0.2539683	0.7145944
Random Forest	0.5	0.00000	1.0000000	NaN	0.7460317	0.2539683	0.6964761

- **Random Forest was used as the nonparametric model because it can capture nonlinear patterns and interactions automatically. At the default 0.50 threshold, it predicted every woman as a non-fracture, giving 0% sensitivity and 100% specificity.**
- **The highest predicted fracture probability was only 0.496, so the 0.50 cutoff was too high. Its AUROC was 0.699, showing some ability to rank fracture risk, but lower thresholds were needed to make the model useful.**

Altering the Thresholds of the Final Models

Model	Threshold	Sensitivity	Specificity	PPV	NPV	Prevalence	AUROC
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Additive Logistic	0.21	0.68750	0.6702128	0.4150943	0.8630137	0.2539683	0.7232380
Complex Logistic	0.22	0.65625	0.6808511	0.4117647	0.8533333	0.2539683	0.7175864
Complex Categorical BMI Logistic	0.21	0.65625	0.6489362	0.3888889	0.8472222	0.2539683	0.7177527
Complex Categorical BMI Logistic with Changed Groups	0.21	0.65625	0.5638298	0.3387097	0.8281250	0.2539683	0.6662234
LDA	0.20	0.68750	0.6595745	0.4074074	0.8611111	0.2539683	0.7205785
QDA	0.23	0.65625	0.6702128	0.4038462	0.8513514	0.2539683	0.7145944
Random Forest	0.25	0.65625	0.7021277	0.4285714	0.8571429	0.2539683	0.6994681

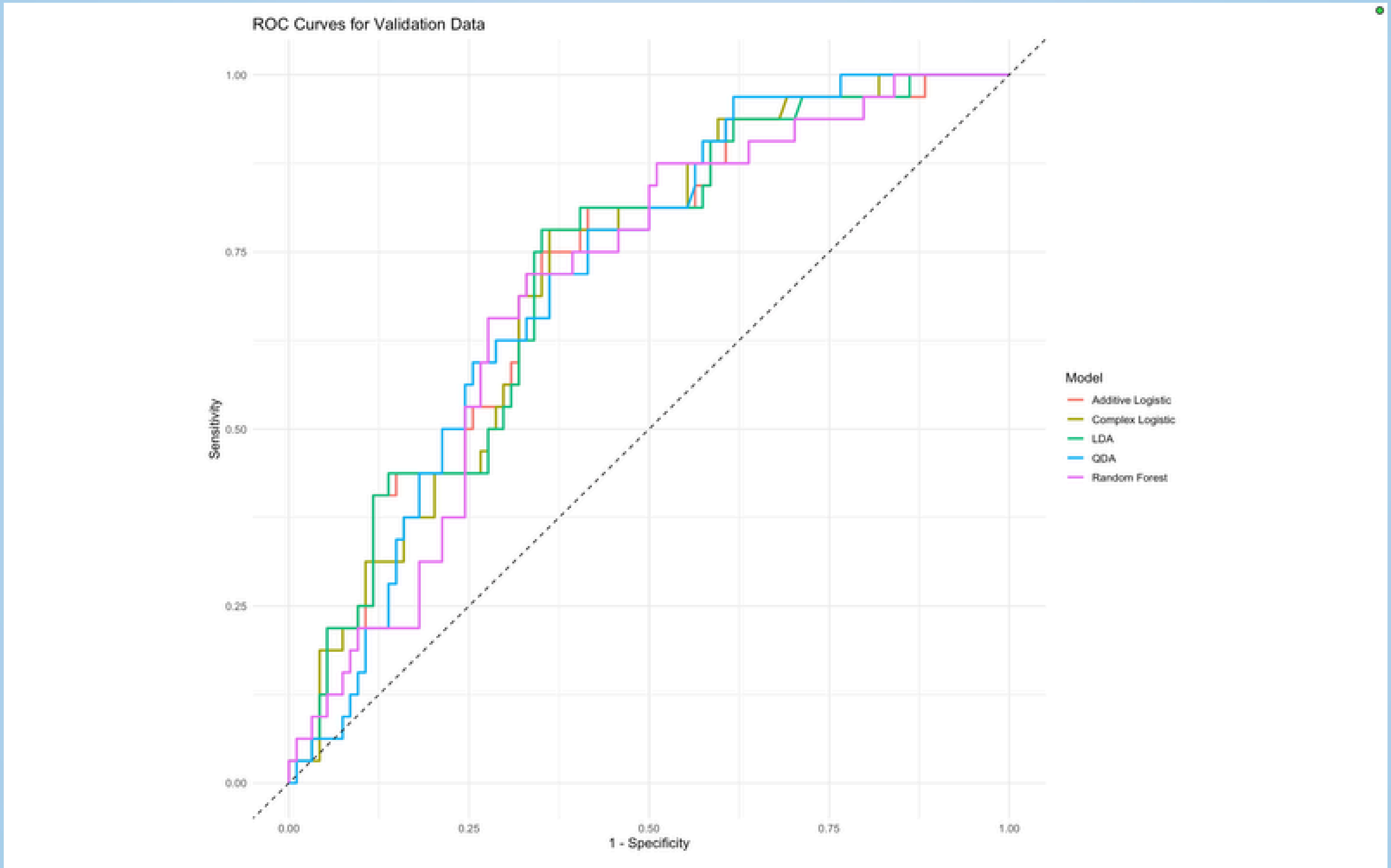
- The 0.50 threshold gave very low sensitivity for most models, so lower model-specific thresholds were explored to better identify women at risk of fracture. Thresholds were chosen to keep sensitivity around 65%–75% while maintaining reasonable specificity.

- After adjustment, the models became more balanced. The additive logistic model had 68.8% sensitivity and 67.0% specificity, while Random Forest had slightly lower sensitivity but the highest specificity at 70.2%.

- PPV remained modest, but NPV was generally above 82%, so the models were better at ruling out fracture than confirming it. The additive logistic model still had the highest AUROC at 0.723.

- Because thresholds were chosen using the validation data, they should be viewed as practical cutoffs for this analysis rather than fully validated optimal thresholds. Overall, lowering the thresholds improved sensitivity and produced a better tradeoff between detecting fractures and avoiding false positives.

ROC Curve Plot



Model Recommended for Predictive Accuracy

$$\text{fracture} \sim \beta 0 + \beta 1(\text{age}) + \beta 2(\text{height}) + \beta 3(\text{priorfrac}) + \beta 4(\text{momfrac}) + \beta 5(\text{armassist}) + \beta 6(\text{smoke}) + \beta 7(\text{premeno}) + \beta 8(\text{raterisk}) + \beta 9(\text{bonemed})$$

- **Recommended model: Additive Logistic Regression**
- **AUROC: 0.723**
- **Sensitivity: 68.8% at threshold 0.21**
- **Reason: Best overall balance of performance and interpretability**

Conclusion

Objective 1: Key predictors were age, height, prior fracture, and maternal fracture, with both fracture-history variables roughly doubling the odds of first-year fracture.

Objective 2: No complex or alternative model outperformed the additive logistic regression. LDA showed similar discrimination, while QDA and Random Forest shifted sensitivity–specificity trade-offs but had lower AUROC. This likely reflects the dataset's small size, imbalance, and modest signal, where simpler models tend to perform best.

BMI Experiments: Interaction and categorical-BMI models did not improve validation performance, indicating that added complexity was unnecessary.

Overall: The additive logistic model offered the best balance of accuracy and interpretability.

Practical Use: Predictions should be limited to women similar to those in the dataset, with consistent variable definitions and predictor ranges.

Future Work: Thresholds were not externally validated, so independent validation is needed before any clinical use.

Thank You!

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